

Computational Viability Certification under Incomplete Observation

Supplementary Material: Complete Proofs and the Verification Record

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S1. Complete proofs

Throughout, labels are $a = (\theta, i, t) \in \mathcal{S} = \Theta \times \{1, \dots, s\} \times [0, T]$, k_a , β_a , $\mathcal{F}_a$, $J_{\mathcal{I}}(T)$, ϕ_U , $\Gamma_{\mathcal{I}}$, $\rho_{\mathcal{H},G}$, and the program (5) are as in the main text.

Proof of Proposition 1 (robust rows). The linear solution formula gives, for $t \leq T$,

$$c_i^\top x_\theta(t) = c_i^\top X_\theta(t, 0)x_0 + \int_0^t c_i^\top X_\theta(t, \sigma)g_\theta(\sigma) d\sigma + \int_0^t k_a(\sigma)^\top u(\sigma) d\sigma + \int_0^t c_i^\top X_\theta(t, \sigma)E_\theta(\sigma)d(\sigma) d\sigma.$$

The initial-state term is bounded over $x_0 \in P_\theta$ by the support value $h_{P_\theta}(X_\theta(t, 0)^\top c_i)$, and the disturbance term, maximized over all measurable $d(\cdot)$ with values in D_θ , is $\int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma$; for polytopic D_θ a maximizing vertex selection is measurable, so the bound is attained by an admissible disturbance signal. Collecting terms, the constraint $c_i^\top x_\theta(t) \leq b_i$ holds for every (x_0, d) if and only if $\mathcal{F}_a(\pi) \leq 0$, and the separation of the maximizations from the control term is exactly that the control signal in a mode does not depend on x_0 or d .

Proof of Proposition 2 (value and nonviability). The admissible signal space is a finite product of the sets $L^\infty(I_r; U)$; because U is compact and convex these are weak-* compact by the Banach–Alaoglu theorem. The safety functionals are affine and weak-* continuous because their kernels k_a lie in L^1 . Their supremum over the finitely many labels is therefore weak-* lower semicontinuous and attains its minimum on the policy space: $J_{\mathcal{I}}(T)$ is well defined. Safety of some policy is $\max_a \mathcal{F}_a(\pi) \leq 0$, so $B_0 \in \mathcal{K}_{\mathcal{I}}^T$ iff $J_{\mathcal{I}}(T) \leq 0$; since safety at the margin is viability (constraints closed) and $J_{\mathcal{I}}(T) = 0$ admits safe policies, nonviability is $J_{\mathcal{I}}(T) > 0$. The strict inequality is thus a consequence of compactness and closedness, not an added assumption.

Proof of Theorem 4 (continuous-to-finite). *Step 1 (control-moment inequality).* For a block J_j and any measurable u with values in U ,

$$\left(\int_{J_j} k_a \right)^\top \frac{1}{|J_j|} \int_{J_j} u - \int_{J_j} k_a^\top u = \int_{J_j} (\bar{k}_{a,\mathcal{H}} - k_a)^\top u \leq \int_{J_j} h_U(\bar{k}_{a,\mathcal{H}} - k_a),$$

using $\sup_{v \in U} w^\top v = h_U(w)$. Summing over blocks proves (3); the averages remain compatible with the information structure because the mesh contains all sensing times, so common-history controls use the same block variable. For the bound, choose a center u_c of U ; since $\int_{J_j} (k - \bar{k}) = 0$, the center cancels and $e_{a,\mathcal{H}} \leq R_U \|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1}$. A bounded-variation function satisfies $\|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1} \leq h_u \text{TV}(k_a)$, whence $e_{a,\mathcal{H}} \leq R_U V_U h_u$. The pointwise bound behind it: for every $\sigma \in J$, $\|\bar{k} - k(\sigma)\| = \left\| \frac{1}{|J|} \int_J (k(\tau) - k(\sigma)) d\tau \right\| \leq \text{TV}(k; J)$, the norm of any increment being at most the total variation; integrate over J and sum. The same estimate bounds the reflected integrand $\int h_U(k_a - \bar{k})$ (center cancellation is symmetric), which is what Step 3 uses.

Step 2 (lower bound of (8)). Take any admissible continuous-time policy and form its block averages, which lie in U and satisfy (3). For every grid label,

$$A_a v - \beta_a^+ - \bar{e}_a \leq \int_0^T k_a^\top u_\theta^\pi - \beta_a = \mathcal{F}_a(\pi),$$

so the finite program attains a value no larger than the maximum robust violation of that policy (the display needs only the one-sided bound from (3), $A_a v - \int k_a^\top u^\pi \leq \bar{e}_a$, and $\beta_a^+ \geq \beta_a$. The reflected error can be larger than $\bar{e}_a$, so an absolute-value bound is not inferred). Minimizing over policies gives $\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T)$ — the block-moment vector of every admissible policy is feasible for the finite program, at objective $\max_a \mathcal{F}_a(\pi)$.

Step 3 (upper bound of (8)). Take an optimizer v^* of (5) and implement the corresponding piecewise-constant information-adapted control, for which $\int_0^T k_a^\top u_\theta^{v^*} = A_a v^*$ — exactly, with no moment error, because the implemented control is constant on each block; the implementation price is paid only through the grid labels

versus all times below. At grid times, $\mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + \bar{e}_a + \delta_\beta$; the Lipschitz hypothesis (6) extends the bound to every time ($|\mathcal{F}_a(v^*) - \mathcal{F}_{\hat{a}}(v^*)| \leq L|t - \hat{t}| \leq Lh_t$ for any time label within h_t of the grid label $\hat{a}$), giving $\max_a \mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + \max_{\hat{a} \in \mathcal{S}_G} \bar{e}_{\hat{a}} + \delta_\beta + Lh_t$. Since $J_{\mathcal{I}}(T)$ minimizes over all admissible policies, the upper bound follows. Together with (4) this gives the sharpened form; the uniform choice $\bar{e} \equiv R_U V_U h_u$ — valid by the moment-error bound (2) whenever $\text{TV}(k_a) \leq V_U$ — recovers the announced special case. The asymmetry — averaging arbitrary controls proves the lower bound, implementing piecewise-constant controls proves the upper bound — is what makes naïve infeasibility-by-discretization arguments unsound.

Step 4 (sound finite certificates). Write the block input constraints as $F_{\text{blk}} v \leq f_{\text{blk}}$. The program's dual is

$$\max_{\lambda, \mu} -\lambda^\top (\beta^+ + \bar{e}) - \mu^\top f_{\text{blk}} \quad \text{s.t.} \quad A^\top \lambda + F_{\text{blk}}^\top \mu = 0, \quad \lambda \geq 0, \quad \mathbf{1}^\top \lambda = 1, \quad \mu \geq 0, \quad (12)$$

so a positive dual objective certifies infeasibility of $\{Av \leq \beta^+ + \bar{e}, v \in U^Q\}$. By Step 1 every genuinely safe measurable controller would generate a feasible v ; the finite infeasibility certificate therefore excludes every such controller. Moreover the weighted continuous-time violation is at least the dual objective: a normalized certificate with margin $\gamma > 0$ guarantees an actual facet violation of at least γ in one stored test (Remark 1 of the main text).

Step 5 (completeness). If the system is nonviable, let $\gamma = J_{\mathcal{I}}(T) > 0$ and choose meshes and stored scenarios making the error in (8) — including the stored-scenario widths δ_{init} and δ_{num} , which near-maximizing stores contribute and completeness must drive to zero — smaller than γ ; then $\rho_{\mathcal{H},G} > 0$ and Step 4's finite LP duality yields a positive certificate, proving (iii). No belief-state recursion is used.

Step 5b (exact atomic dual representation, proof of (9)). Write Π for the policy space and $\mathcal{S}$ for the label set.

(a) *Policy compactness.* Each factor $\{u \in L^\infty(I_r; U)\}$ is weak-* closed — U is a finite intersection of halfspaces $f_i^\top v \leq g_i$, and for $f_i \mathbf{1}_{I_r} \in L^1$ the set $\{u : f_i^\top u(\sigma) \leq g_i \text{ a.e.}\}$ is weak-* closed — and weak-* bounded; Banach–Alaoglu makes it weak-* compact and convex. The finite product Π is therefore weak-* compact, convex, and metrizable (each L^1 is separable, so bounded weak-* sets are metrizable).

(b) *Joint continuity of the label-policy pairing.* Each $\mathcal{F}_a$ is a finite sum of L^1 -pairings against cell signals minus a constant, so $\pi \mapsto \mathcal{F}_a(\pi)$ is weak-* continuous. Under the label-continuity hypothesis ($\|k_a - k_{a'}\|_{L^1} \rightarrow 0$, $|\beta_a - \beta_{a'}| \rightarrow 0$; the L^1 form is the operative one — the sup-norm form is false for the step kernels, which jump at $\sigma = t$ for every t),

$$\sup_{\pi} |\mathcal{F}_a(\pi) - \mathcal{F}_{a'}(\pi)| \leq M_U \|k_a - k_{a'}\|_{L^1} + |\beta_a - \beta_{a'}| \rightarrow 0,$$

with $M_U := \sup_{v \in U} \|v\|$ (Hölder: the policies take values in U , so each pairing against an L^1 kernel difference is bounded by M_U times its L^1 norm), uniformly in π ; with separate continuity this gives joint continuity of $(a, \pi) \mapsto \mathcal{F}_a(\pi)$ on the compact $\mathcal{S} \times \Pi$, in particular boundedness.

(c) *Mixed extension and Sion.* Let $\mathcal{P}(\mathcal{S})$ be the Borel probability measures with the weak-* topology: by the Riesz representation theorem it is compact, convex, and metrizable. The bilinear $G(\pi, \mu) = \int \mathcal{F}_a(\pi) d\mu(a)$ is affine in each variable — hence simultaneously quasi-convex and quasi-concave — and continuous in each variable (π -continuity by dominated convergence against the fixed finite μ ; μ -continuity by pairing against the continuous $a \mapsto \mathcal{F}_a(\pi)$). Sion's minimax theorem (both spaces compact convex) gives $\min_{\pi} \max_{\mu} G = \max_{\mu} \min_{\pi} G$ with the outer maximum attained; and $\max_{\mu} G(\pi, \cdot) = \max_a \mathcal{F}_a(\pi)$, because the integral of a continuous function over a probability measure lies within its min–max range while the Dirac measures attain both ends. Hence $J_{\mathcal{I}}(T) = \max_{\mu} \min_{\pi} G$.

(d) *Continuity of the value function.* $g(\mu) := \min_{\pi} G(\pi, \mu)$ is weak-* continuous: by (b) the family $\{a \mapsto \mathcal{F}_a(\pi) : \pi \in \Pi\}$ is uniformly bounded and uniformly equicontinuous in $C(\mathcal{S})$, so Arzelà–Ascoli makes its closure compact, and $\mu_n \rightarrow \mu$ weak-* forces $\sup_{\pi} |G(\pi, \mu_n) - G(\pi, \mu)| \rightarrow 0$; evaluating at near-minimizing π 's gives $g(\mu_n) \rightarrow g(\mu)$.

(e) *Atomic approximation.* Finitely supported measures are weak-* dense in $\mathcal{P}(\mathcal{S})$: quantize $\mathcal{S}$ into finitely many Borel sets of diameter below $1/n$, keep one representative point per set, and give it the set's μ -mass; the resulting finitely supported measures converge to μ . Taking $\mu_n \rightarrow \mu^*$, the maximizer of (c), (d) gives $g(\mu_n) \rightarrow g(\mu^*) = J_{\mathcal{I}}(T)$. Hence

$$J_{\mathcal{I}}(T) = \sup \left\{ \min_{\pi} \sum_{\ell} \lambda_{\ell} \mathcal{F}_{a_{\ell}}(\pi) : q \in \mathbb{N}, a_{\ell} \in \mathcal{S}, \lambda \in \Delta_q \right\},$$

which is (9) once the inner minimum is identified with $\Gamma_{\mathcal{I}}$.

(f) *The Γ representation.* For $\mu = \sum_{\ell} \lambda_{\ell} \delta_{a_{\ell}}$ the inner minimum decomposes over information cells (r, C) : it is the infimum over $u_{r,C} \in L^\infty(I_r; U)$ of $\int_{I_r} g_{r,C}(\sigma)^\top u_{r,C}(\sigma) d\sigma$ with $g_{r,C} = \sum_{\ell: \theta_{\ell} \in C} \lambda_{\ell} k_{a_{\ell}}$. The inequality $\leq \int \phi_U(g_{r,C})$ is the pointwise bound; the converse is measurable selection — for a.e. σ the set of minimizers of $v \mapsto g_{r,C}(\sigma)^\top v$ over the polytope U is a nonempty face, the lexicographically first minimizing vertex in a fixed

vertex order is measurable in σ , and selecting it attains the pointwise minimum a.e., so the infimum equals $\int_{I_r} \phi_U(g_{r,C})$. Summed over cells, $\min_{\pi} \sum_{\ell} \lambda_{\ell} \mathcal{F}_{a_{\ell}}(\pi) = \Gamma_{\mathcal{I}}(\lambda, a)$ as defined in (1).

(g) *Nonviability equivalence.* If $J_{\mathcal{I}}(T) > 0$, (e) supplies a finitely supported μ with $g(\mu) > J_{\mathcal{I}}(T)/2 > 0$: a positive finite atomic certificate. Conversely $\Gamma_{\mathcal{I}}(\lambda, a) = \min_{\pi} \sum_{\ell} \lambda_{\ell} \mathcal{F}_{a_{\ell}}(\pi) \leq \min_{\pi} \max_a \mathcal{F}_a(\pi) = J_{\mathcal{I}}(T)$ because the maximum bounds the λ -weighted average. $\square$

Step 6 (sparsity). Factor the finite row map through its rank, $Av = L(Rv)$, $Rv \in \mathbb{R}^r$; the attainable moment set $\mathcal{M} = \{Rv : v \in U^Q\}$ is compact and convex, and each safety row defines a halfspace in $\mathbb{R}^r$. If the common feasible set is empty, Helly's theorem in $\mathbb{R}^r$ gives an obstruction involving at most $r + 1$ safety halfspaces, and finite duality on that subproblem yields a sparse positive certificate with the same objective: $r + 1 \leq mQ + 1$ labels. To preserve a prescribed margin $\gamma_0 < \rho_{\mathcal{H},G}$, apply the argument to the still-infeasible shifted constraints $A_a v \leq \beta_a^+ + \bar{e}_a + \gamma_0$. The continuous finite-rank case follows identically on the compact attainable moment body of the lifted kernels, giving $r_{\mathcal{I}} + 1$. $\square$

Proof of Proposition 3 (arbitrarily large minimal obstructions). With the construction of the main text and $w_j = \int_{I_j} u$: mode $j \leq q$ ends the interval at $x_j = w_j - 1 \geq 0$, so safety of the blind window requires $w_j \geq 1$; mode $q + 1$ ends at $q - \varepsilon - \sum_j w_j$, so safety requires $\sum_j w_j \leq q - \varepsilon$. Summing the q lower bounds against the upper bound gives $0 \leq -\varepsilon$: a contradiction, so the full belief is nonviable. Removing mode $q + 1$, $u \equiv 1$ gives $w_j = 1$ and $x_j = 0$: safe. Removing mode $j \leq q$, $u = 0$ on I_j and $u = 1$ elsewhere gives $x_{q+1} = (q - 1) - (q - \varepsilon) + 1 = \varepsilon \geq 0$ at the end and safety on the remaining branch: every proper subbelief is viable. Equal weights $\lambda_j = 1/(q+1)$ on the $q + 1$ endpoint rows give pooled pre-kernels $\sum_j \lambda_j k_j = 0$ on every interval where the modes agree and a margin of $\varepsilon/(q+1)$ at the horizon. The kernel rank: the endpoint rows are $\mathbf{1}_{I_j}$ -weighted copies of the single input direction with one linear relation (their weighted sum), so the rank of the safety-row matrix is exactly q , and Helly-type sparsity in that rank requires $q + 1$ witnesses — no bound in the input dimension $m = 1$ exists. $\square$

Derivation of the moduli from primitive data. The hypothesis constants of Theorem 4 are not free assumptions; under the standing bounds they are computed from the data. Let $\|A_{\theta}\|_{\infty} \leq \bar{A}$, $\|B_{\theta}\|_{\infty} \leq \bar{B}$, $\|E_{\theta}\|_{\infty} \leq \bar{E}$, $\|g_{\theta}\|_{\infty} \leq \bar{g}$, $\|c_i\| \leq \bar{c}$ on $[0, T]$; $U \subset c_U + R_U \mathbb{B}$, $D_{\theta} \subset R_D \mathbb{B}$, $P_{\theta} \subset R_P \mathbb{B}$; $\text{TV}(B_{\theta}) \leq V_B$, $\text{TV}(E_{\theta}) \leq V_E$; and $M_X := e^{AT}$, so that $\|X_{\theta}(t, \sigma)\| \leq M_X$ (Grönwall on the transition matrix) and $\frac{\partial}{\partial \sigma} X_{\theta}(t, \sigma) = -X_{\theta}(t, \sigma) A_{\theta}(\sigma)$.

Total variation of the kernels. For $k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_{\theta}(\sigma)^{\top} X_{\theta}(t, \sigma)^{\top} c_i$: on $[0, t]$ the product rule for functions of bounded variation gives

$$\text{TV}(k_a; [0, t]) \leq \sup_{\sigma} \|X_{\theta}(t, \sigma)^{\top} c_i\| \text{TV}(B_{\theta}) + \sup_{\sigma} \|B_{\theta}(\sigma)\| \int_0^t \|X_{\theta}(t, \sigma) A_{\theta}(\sigma)^{\top} c_i\| d\sigma \leq \bar{c}(M_X V_B + \bar{B} M_X \bar{A} T),$$

and the drop to zero at $\sigma = t$ adds at most $\|B_{\theta}(t)^{\top} c_i\| \leq \bar{B} \bar{c}$ (using $X_{\theta}(t, t) = I$). Hence (7) holds with $V_U = \bar{c}(\bar{B} + M_X V_B + \bar{B} M_X \bar{A} T)$, and identically for the disturbance kernel with $V_D = \bar{c}(\bar{E} + M_X V_E + \bar{E} M_X \bar{A} T)$.

L^1 label-continuity modulus. The same bounds give the L^1 continuity in the label time that the proof of (9) consumes (the sup-norm form is false here: the step kernels jump at $\sigma = t$ for every t). For $t \geq t'$, Grönwall on $X_{\theta}(t, \sigma) - X_{\theta}(t', \sigma) = \int_{t'}^t A_{\theta}(r) X_{\theta}(r, \sigma) dr$ gives $\|X_{\theta}(t, \sigma) - X_{\theta}(t', \sigma)\| \leq M_X \bar{A} |t - t'|$ uniformly in σ , so with $k_a = \mathbf{1}_{\{\sigma \leq t\}} B_{\theta}(\sigma)^{\top} X_{\theta}(t, \sigma)^{\top} c_i$,

$$\|k_t - k_{t'}\|_{L^1} \leq \bar{B} \bar{c} |t - t'| + \int_0^{t'} \bar{B} \bar{c} M_X \bar{A} |t - t'| d\sigma \leq \bar{B} \bar{c} (1 + M_X \bar{A} T) |t - t'|,$$

and $|\beta_t - \beta_{t'}| \leq L^{\text{tr}} |t - t'|$ by the trajectory Lipschitz bound below: the label map is L^1 -Lipschitz in t with constant $\bar{B} \bar{c} (1 + M_X \bar{A} T) + L^{\text{tr}}$, which is the form the label-continuity hypothesis consumes.

Uniform time-Lipschitz constant. For a trajectory x of mode θ : $\|x(t)\| \leq M_X (R_P + T(\bar{g} + \bar{B}(R_U + \|c_U\|)) + \bar{E} R_D)$ by Grönwall, and

$$\left| \frac{d}{dt} c_i^{\top} x(t) \right| \leq \bar{c} (\bar{A} \|x(t)\| + \bar{g} + \bar{B}(R_U + \|c_U\|) + \bar{E} R_D) \leq L^{\text{tr}},$$

with $L^{\text{tr}} := \bar{c} \bar{A} M_X R_P + \bar{c} (1 + \bar{A} M_X T) (\bar{g} + \bar{B}(R_U + \|c_U\|) + \bar{E} R_D)$. Each $\mathcal{F}_{(\theta, i, t)}(\pi)$ is a supremum over (x_0, d) of the L^{tr} -Lipschitz functions $t \mapsto c_i^{\top} x(t) - b_i$, and a pointwise supremum of L -Lipschitz functions is L -Lipschitz; hence (6) holds with $L = L^{\text{tr}}$. For the three-branch instance the closed form applies (with $\bar{A} = 1$, $M_X = e^{6/5}$, $R_P = 34/25$, $\bar{B} = 1$, $R_U = 1$); the direct kernel computation of Section 6 gives the sharper instance constants $V_U = 6/5$, $L = 11/5$, $R_U = 1$, $\delta_{\beta} = 0$ used in the sandwich reading of the campaign.

Proof of the erosion identity. ϕ_U is concave and positively homogeneous; for any decomposition, $\phi_U(\sum_j g_j) \geq \sum_j \phi_U(g_j)$ because each $\phi_U(g_j) = \min_{v \in U} g_j^\top v$ is attained at a common minimizer for the sum while separate minima only lower the total. Consequently the same witness family, evaluated on the coarse partition, dominates its evaluation on the refinement, $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, and the exact erosion is the displayed integral (12). For $U = [-\bar{u}, \bar{u}]^m$, $\phi_U(g) = -\bar{u}\|g\|_1$, giving the stated ℓ^1 form. The 3,240-pair verification for the hexagonal U is reported in S2. $\square$

Proof of the oracle-recourse certificate (13). The oracle relaxation replaces the true post- τ policy class by the larger class that may assign each stored scenario its own measurable control. The certificate (13) is the atomic margin (1) evaluated in that enlarged class: before τ the pooled kernel term is $\phi_U(\sum_\ell \lambda_\ell k_\ell)$, after τ each scenario ℓ contributes $\phi_U(\lambda_\ell k_\ell)$ with its own margin β_ℓ computed from the stored scenario. A positive margin certifies infeasibility in the enlarged class by Propositions 1–2; every policy of the true class is feasible for the enlarged one, so the verdict transfers downward: no actual output-feedback policy is safe. Soundness is thus immediate; completeness fails because the oracle class is strictly larger.

S2. The verification record

All numbers in the main text are re-derived in exact rational arithmetic (Python `fractions`; no floating point in any witness computation; floats only inside the solver call, whose output is compared against the exact value only within solver tolerance). Two scripts carry the record, both at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`):

- `paper2c_lp_campaign.py` — the solver-assisted, exactly certified mesh campaign: builds program (5) at each mesh $h \in \{0.2, 0.1, 0.05, 0.02, 0.01\}$, verifies the dimension identities ($mQ+1$ variables; $Ms(N_t+1) + p_U Q$ rows), solves with HiGHS, and certifies optimality of $\rho(h) = \max\{0, \frac{3}{50} - Th/4\}$ by the exact primal witness (pre-observation blocks 0; post-observation blocks of mode j at $-n_j$; every row checked) and the exact dual witness ($\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$, $\mu_{j,k} = \lambda_j a_k \eta_j$; stationarity and objective $-\rho(h)$ per block). Also re-derives $\Gamma(\tau) = \tau - \frac{7}{50}$, $\tau_{\max} = \frac{7}{50}$, the $h = \frac{1}{10}$ Farkas contradiction $-\frac{3}{100}$, and the pair-viability positions $1.9752 = \frac{12345}{6250}$ and $1.9352 = \frac{2419}{1250}$, attained by the recorded pairwise policies (blind-window hold $u = 0$ resp. $u = n_1$, then the two-phase brake $-\frac{2}{5}n_j$ for $\frac{1}{5}$ and $-n_j$ to rest; Section 6 of the main text). The witness pair is mesh-generic on the aligned family: re-running it exactly at the sixth aligned mesh $h = \frac{1}{15}$ ($Q = 48$, 687 inequalities) certifies $\rho = \frac{1}{25}$ there, extending the law’s verified domain beyond the five campaign meshes. Current run: 9/9 checks pass; solver agreement at most 4.2×10^{-17} on all five meshes.
- `paper2_nonstandard_verification.py` — the 51-check exact-arithmetic record for the predecessor material: the three-branch instance values, the $m+1$ construction for all tested q , the ϕ_U superadditivity on 3,240 rational pairs, the error-bound checks, and the counterexamples. Current run: 51/51 checks pass.

The mesh table of the main text is regenerated verbatim by the first script; the reported iteration counts and wall-clock times are measured values from the run recorded here (scipy HiGHS, dual simplex), not estimates.